

CMAN

$$10T \text{ FLOPs/sec} = 10,000 \text{ GFLOPs/sec}$$

perf GFLOPs/sec

RIDGE POINT (31.25, 10T)

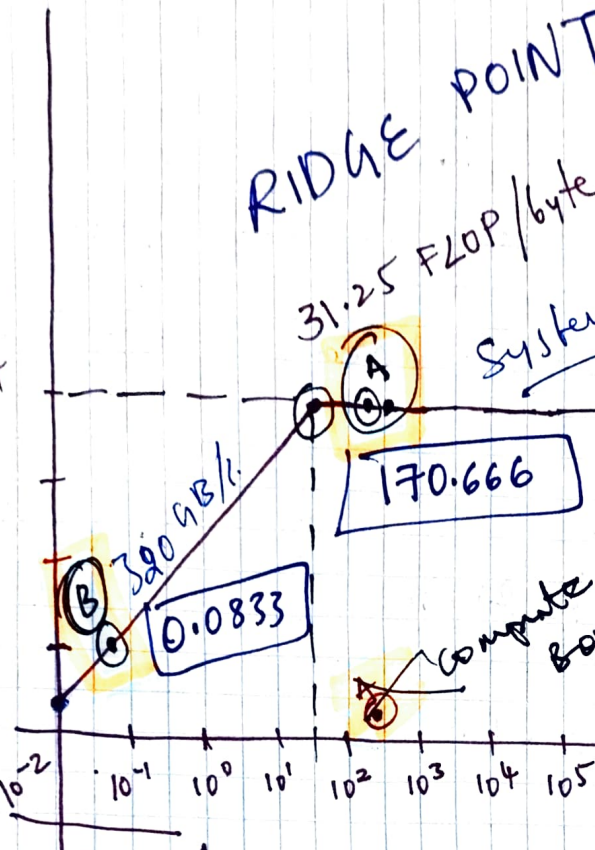
31.25 FLOP/byte.

System Compute Band.

10T FLOPS

reach compute

10⁴
10³
10²
10¹
10⁰
10⁻¹
10⁻²



The kernel points A, B are plotted wrong. We only need "A" to plot, not the Y coord.

Arth Intensity FLOP/Byte.

Y-intercept

$$= \log_{10}(320) = 2.505$$

Extremely Memory Bound.

① FLOP

① FLOP

② B

③ Bytes FP32

FP8

2B

FP16

0.083,
0.333

↓ precision

1. 1024×1024 GEMM

$$\text{FLOPs} = 2 \times (1024)^3$$

$$2.1474 \text{ GFLOPs}$$

$$= 1073741824$$

$$= 2147483648$$

$$\text{Bytes Transferred} = 3N^2 \times 4$$

$$= 3(1024)^2 \times 4$$

$$= 3(1048576) \times 4$$

$$\text{Bytes Transferred} = 12582912$$

$$\text{Arth. I} = 170.6666 \text{ FLOP/B}$$

Kernel A:

$$\begin{aligned} \text{GFLOPs} &= 2.1474 \\ \text{AI} &= 170.666 \end{aligned}$$

~~Kernel B:~~

COMPUTE BOUND

$$N/6$$

ATTAINABLE

GFLOPs

$$= 10 \text{ TFLOPs/s}$$

~~How DID YOU GET THIS~~

kernel B:

① Bytes fetched = $(2 \times 4194304) \times 4$
= 33554432

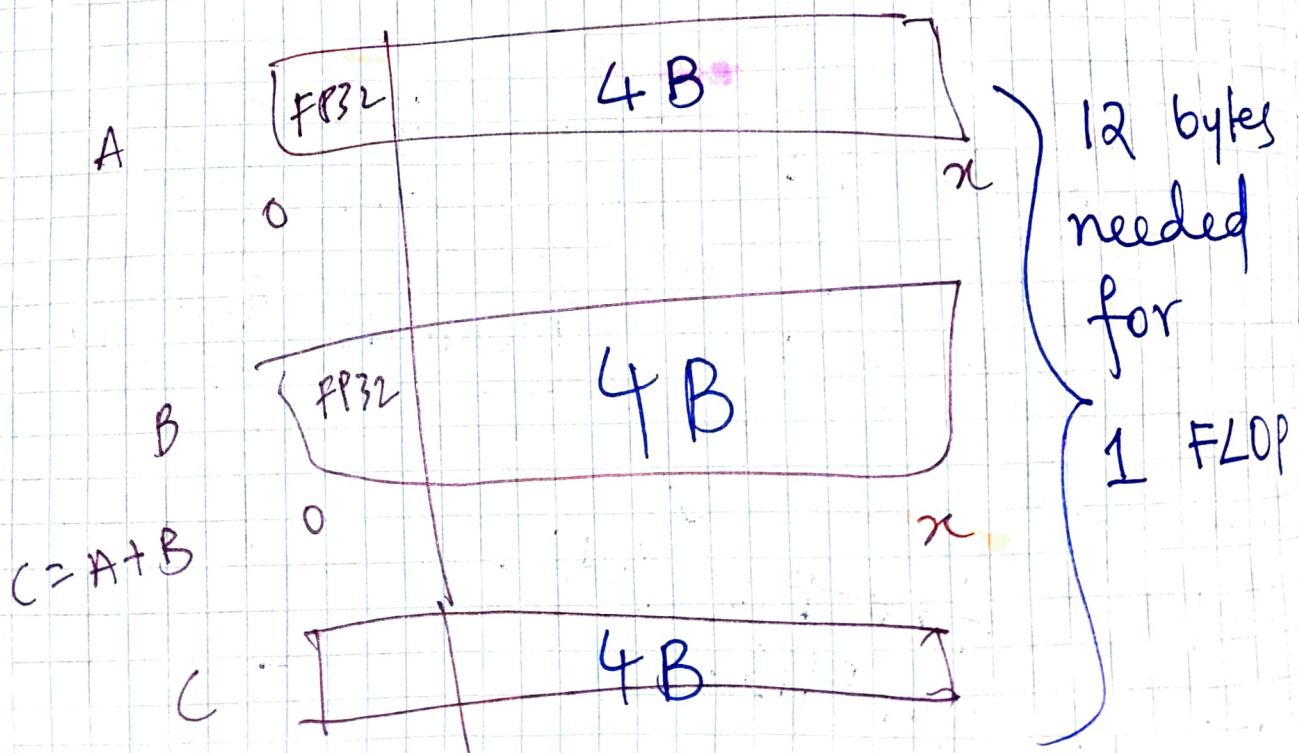
② Bytes written = 4194304×4
= 16777216

Total Bytes = ① + ②
= 50331648

FLOPS = 4194304

Giga

=



$$N = 4194304$$

4 Bytes.

$$3 \times N \times 4$$

$$= 12 \times N$$

$$\text{FLOPS} = N.$$

Attainable
GFLOPS

$$= 0.083333$$

$$\times 320 \times 10^9$$

$$= 26.67$$

GFLOPS.

$$\text{GFLOPS} = 0.004194304$$

$$\text{AI} = \frac{4194304}{50331648} =$$

$$0.083333$$

Kernel B is in memory bound region.

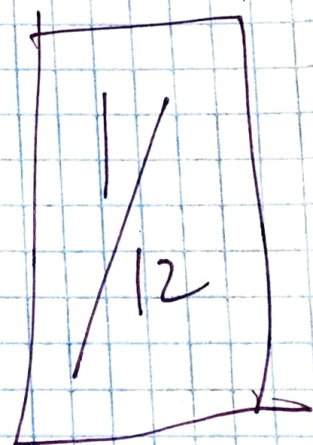
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(a) Kernel A $\rightarrow$ Compute Bound

(b) K B $\rightarrow$ Mem Bound

(c) for A $\rightarrow$ More TFLOPS

for B $\rightarrow$ More Mem Bandwidth.



For kernel A, increase the Arithmetic units so that the problem can operate at much higher peak performance.

~~For~~ Since, the kernel A is operating on well over the ridge point on right side, it has enough data to work on, but not enough compute units.

For kernel B, the AI is $1/2$ so it needs very high bandwidth memory, using wider data BUS, HBM and operating at higher freq DRAM would improve the ~~opt~~ utilisation.